Argument 2: When we change the transformer architecture in a way that shifts when induction heads form or whether they can form, the dramatic improvement in in-context learning shifts in a precisely matching way.

STRENGTH OF ARGUMENT FOR SUB-CLAIMS

	Small Attention-Only	Small with MLPs	Large Models
Contributes Some	Medium, Interventional	Medium, Interventional	Weak, Interventional
Contributes Majority	Medium, Interventional	Medium, Interventional	Weak, Interventional

One thing that falls short about Argument 1 is that we're just observing induction heads and in-context learning co-vary; like any observational study, it is not as convincing as if we actively changed one thing and measured what happened to the other. In this section we do a more "interventional" experiment, in which we change the model architecture in a way that makes induction heads easier to form, and observe the effect on in-context learning. The change makes a bigger difference for smaller models so is more convincing there, but also carries some amount of weight for larger models (see table above).

To design our experiment, we start with the observation (noted in the previous section) that the phase change and the corresponding improvement in in-context learning only occurs in transformers with more than one layer. This is what we'd predict if induction heads were the mechanism for the majority of in-context learning: induction heads require a composition of attention heads, which is only possible with two or more layers.[15]

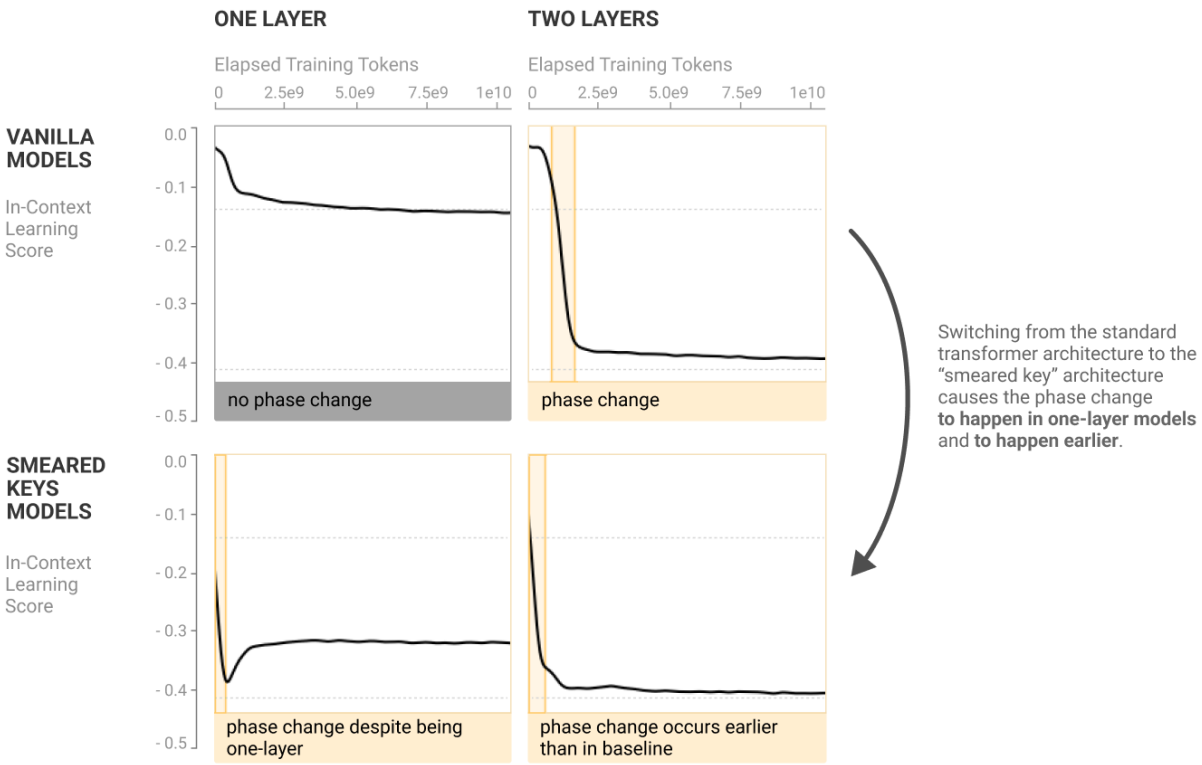
Of course, the observation about one-layer models is pretty weak evidence by itself. (One could imagine one-layer models being different from models with more layers in all sorts of ways!) But it suggests a more general line of attack. If induction heads are the mechanism behind the large improvement in in-context learning, that makes predictions about the minimum architectural requirements in order to achieve the observed improvement. For a standard transformer, the important thing is to have two attention layers. But that's only because the key vectors need to be a function of the attended token and the token before it.

We define a "smeared key" architecture with a very simple modification that should make it easy for transformers of any depth to express induction heads. In our modified models, for each head h, we introduce a trainable real parameter α^h which we use as $\sigma(\alpha^h) \in [0, 1]$ to interpolate between the key for the current token and previous token[16] :

$$k_j^h \;=\; \sigma(\alpha^h) k_j^h \;+\; (1 - \sigma(\alpha^h)) k_{j-1}^h$$

The hypothesis that induction heads are the primary mechanism of in-context learning predicts that the phase change will happen in one-layer models with this change, and perhaps might happen earlier in models with more layers. If they're one of several major contributing factors, we might expect some of the in-context learning improvement to happen early, and the rest at the same time as the original phase change.

The results (figure below) are in line with the predictions: when we use the smeared-key architecture, in-context learning does indeed form for one-layer models (when it didn't before), and it forms earlier for two-layer and larger models. More such results can be seen in the model analysis table.



These plots are an excerpt of the Model Analysis Table; look there to see more analysis of how the smear models compare to their vanilla counterparts.

However, we probably shouldn't make too strong an inference about large models on this evidence. This experiment suggests that induction heads are the *minimal mechanism* for greatly increased in-context learning in transformers. But one could easily imagine that in larger models, this mechanism isn't the whole story, and also this experiment doesn't refute the idea of the mechanism of in-context learning changing over the course of training.

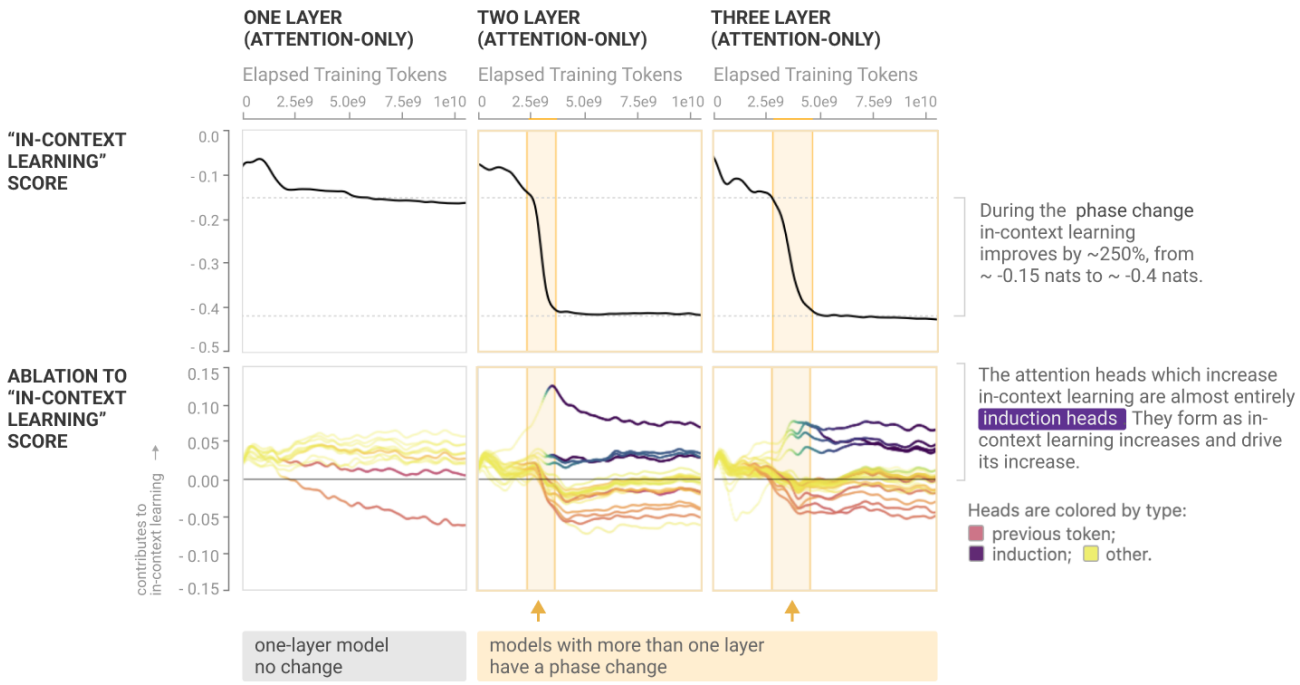
Argument 3: When we directly "knock out" induction heads in small models at test-time, the amount of in-context learning greatly decreases.

STRENGTH OF ARGUMENT FOR SUB-CLAIMS

	Small Attention-Only	Small with MLPs	Large Models
Contributes Some	Strong, Causal	Strong, Causal	
Contributes Majority	Strong, Causal	Medium, Causal	

For the cases they cover, ablations are by far our strongest evidence. The basic argument is that knocking out induction heads decreases the amount of in-context learning we observe in our models. By "knocking out" we mean that we remove a given attention head from the model at test time, doing a forward pass on the transformer without it. (See our methods section for exact details of how the ablation is done.)

The ablations presented below show how attention heads contribute to in-context learning, but we can also do ablations to study how attention heads contribute to the overall behavior change that occurs during the phase change (see the Model Analysis Table).



These plots are a small excerpt of the Model Analysis Table; look there to see the full evidence. See our methods section for exact details of how the ablation is done.